

BAVARIA COP 1809

Common Operating Picture

User Guide

Flight of the Eagle Wargame | U.S. Army War College | 1809 Danube Campaign

PROPRIETARY SOFTWARE

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Version 2.0 — May 2026 | Road Network Edition
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1. Overview

Bavaria COP 1809 is a browser-based Common Operating Picture for the Flight of the Eagle wargame at the U.S. Army War College. It tracks unit positions, calculates road-network march distances, visualizes movement orders, monitors contact between opposing forces, and displays the 1809 Order of Battle hierarchy.

All distance calculations use a hand-built road network matching the period road system shown on the game map. ETAs and march times match what your umpire calculates.

1.1 System Requirements

Requirement	Details
Browser	Google Chrome or Microsoft Edge
Internet	Required on first open to load OpenStreetMap base tiles
Files	All four files in the same folder: index.html, game_map_georef.jpg, alps_map_georef.jpg, rhine_danube_map_georef.jpg
Screen	1280×720 minimum recommended
Online	Live at: https://agatwtest.github.io/Bavaria-1809-COP/

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- Enforcement: 17 U.S.C. § 504 (statutory damages) and § 505 (attorney's fees)

1.3 Opening the Application

- Option 1 — Online: go to <https://agatwtest.github.io/Bavaria-1809-COP/> in Chrome
- Option 2 — Local: download all four files to the same folder, open index.html in Chrome

Note: If period map layers show a broken image icon, confirm all four files are in the same folder.

2. Interface Layout

2.1 Header Bar

Control	Function
Clock Display	Current game DTG: DD HHMM APR (e.g. 16 0600 APR)
-1H / -30M	Step clock backward
+30M / +1H	Step clock forward
Speed Selector	Playback speed: 1×, 5×, 30×, 60×
 PLAY /  PAUSE	Start or stop automatic clock
Map Layer Selector	Toggle: Street Map, Game Map, Alps 1809, Rhine-Danube

2.2 Perspective Toggle

- FRENCH — French units show as blue rectangles, Austrians as red diamonds
- AUSTRIAN — Austrians show as blue rectangles, French as red diamonds

2.3 Map Layers

Layer	Description
Street Map	OpenStreetMap — modern roads and geography
Game Map	Pratzen-style Bavaria 1809 tactical map — georeferenced
Alps 1809	Thiers/Dufour Carte des Alpes — full theater strategic overview
Rhine-Danube	Thiers/Dufour Carte No. 29 — Napoleon's western approach axis

Note: Road distance calculations work regardless of which map layer is displayed. The map is visual only.

3. Road Network Distance Calculator

All distances use a hand-built 1809 road network with Dijkstra's shortest-path algorithm. This is the core feature of Version 2.0 — distances match the game rules, not straight lines.

3.1 Distance Modes

Mode	Description	Use When
 ROAD	Shortest path through the road network	Always — matches game rules
 AS CROW FLIES	Straight-line GPS distance	Comparison only — underestimates march time

3.2 Verified Road Distances

Route	Road Distance	Via
Straubing → Regensburg	44 km	Wörth an der Donau → Barbing
Regensburg → Alkofen	14 km	Direct
Regensburg → Abensberg	32 km	Kelheim
Regensburg → Landshut	70 km	Eggmühl → Schierling → Mellersdorf
Ingolstadt → Landshut	64 km	Mainburg → Pfeffenhausen
Straubing → Passau	93 km	Geiselhöring → Plattling → Deggendorf
Donauwörth → Regensburg	102 km	Neuburg → Ingolstadt → Kelheim
Landshut → Braunau	137 km	Erding → Mühldorf → Burghausen

3.3 March Speeds

Unit Type	Speed	Examples
Army/Corps HQ courier	8 km/hr	Napoleon, Berthier
French/Austrian Cavalry HQ	5 km/hr	Davout HQ, Vandamme HQ
French Cavalry	5 km/hr	Montbrun, Saint-Sulpice, Nansouty
French Line Infantry	4 km/hr	Morand, Friant, Gudin, Demont
Austrian Cavalry	4 km/hr	FML von Kienmayer, Liechtenstein
Austrian Infantry	3 km/hr	Hohenzollern, Rosenberg, Hiller

Note: No terrain modifiers or fatigue applied. Add those manually per umpire rules.

4. PLACE Tab

4.1 Pre-Loaded Starting Positions — 16 APR 1809 at 0600

Unit	Location	Faction
Napoleon	Donauwörth	French — Army HQ
Davout HQ, Morand, Saint-Hilaire, Gudin, Friant, Demont, Montbrun, Saint-Sulpice	Regensburg	French — III Corps
Vandamme HQ, Württemberg Cav, Württemberg Inf, Nansouty	Weihmichl	French — VIII Corps
Hiller	Landshut	Austrian — VI Corps
Archduke Charles HQ, Hohenzollern, Rosenberg, Bellegarde, FML von Kienmayer	Straubing	Austrian — Hauptarmee

4.2 Placing Units

- Dropdown method: select faction → unit → town →  PLACE
- Click map: select unit →  CLICK MAP → click location (snaps to nearest town within 5km)
- Add custom town: + ADD CUSTOM TOWN → type name → click map
- Opacity slider: controls how transparent the period map overlay appears

5. MOVE Tab — Movement Orders

5.1 Issuing a Road Order

- Select  ROAD mode (default)
- Select the unit
- Click + ADD WAYPOINT
- Pick destination — distance, travel time, and ETA calculate automatically via road network
- Route panel shows exact road path: e.g. "via Kelheim → Abensberg"
- Enter depart time for Leg 1 (format: DD HHMM, e.g. 16 0800)
- Legs 2 and 3 auto-cascade from previous ETA
- Click ⚡ ISSUE — dashed route line draws on map along the road

Key point: Road distances are longer than straight-line. Regensburg → Landshut is 70km by road (17.5 hrs at 4 km/hr), not 51km straight line (12.75 hrs). This matches your umpire.

5.2 Canceling a Route

- MOVE tab → select unit → ✕ CLEAR

6. OOB Tab

Full 1809 OOB pre-loaded. Click any unit to select it on the map.

- ↶ RESET 1809 — restore full historical OOB
- ✕ CLEAR ALL — remove all hierarchy
- ASSIGN PARENT — reassign any unit to a new commander

7. INTEL Tab — Contact Detection

Alert	Trigger	Meaning
 ENGAGEMENT	Units within 5 km	Battle imminent
 CONTACT	Units within 25 km	Forces in close proximity
 PROJECTED	Destinations within 20 km	Routes converging

8. Quick Reference

Task	How
Place a unit	PLACE → faction/unit/town →  PLACE
Issue road order	MOVE →  ROAD → unit → + ADD WAYPOINT → town + time →  ISSUE
See road path detail	Waypoint panel shows: "via Kelheim → Abensberg"
Switch perspective	Click FRENCH or AUSTRIAN in perspective bar
Change map layer	Header dropdown
Adjust map opacity	PLACE tab → opacity slider
Run clock	Set speed →  PLAY
Reset OOB	OOB tab → ↶ RESET 1809
See contacts	INTEL tab
Cancel a route	MOVE → unit → ✕ CLEAR

9. Troubleshooting

Problem	Solution
Broken image on period map	All four files must be in the same folder. Do not rename the JPG files.
Map appears black	Check internet — Leaflet loads from Cloudflare CDN on first use.
No road connection between towns	Add an intermediate major town as a waypoint leg.
ETA seems too long	Road distances are longer than straight-line — this is correct and matches the game.
Blank page on GitHub	File too large. Use the four-file version (index.html + three JPGs).
File not loading in Chrome	Rename file to remove spaces and parentheses.

10. Credits and Sources

- John H. Gill, From Abensberg to Znaim: the Franco-Austrian War of 1809 (USAWC monograph, v.2, January 2026)
- Jay Luvaas, Napoleon on the Art of War (The Free Press, 1999)
- Napoleon's Historical Correspondence, Germany and Italy 1809
- Flight of the Eagle Umpire Reference V4
- Thiers/Dufour period maps: Histoire du Consulat et de l'Empire (Paulin, Paris) — public domain, georeferenced by Frank L. Adams III
- OpenStreetMap © OpenStreetMap contributors (ODbL)
- Leaflet.js v1.9.4 © Vladimir Agafonkin — BSD 2-Clause License

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